Zetranova SMS — Product Dossier & Update Log

smswebapp1 · School Management System

Last updated: 2026-07-11 04:48 GMT | **Live:** https://smswebapp1.onrender.com | **Source:** github.com/datatitbit/smswebapp1 | **Build:** v=20260720

This is the living record for smswebapp1. **Part 1** is the one-time foundation — everything a CEO or key team member should know about the product. **Part 2** logs each update on top of Part 1, newest work described in plain language with the date and time it was made.

Part 1 — Fundamentals

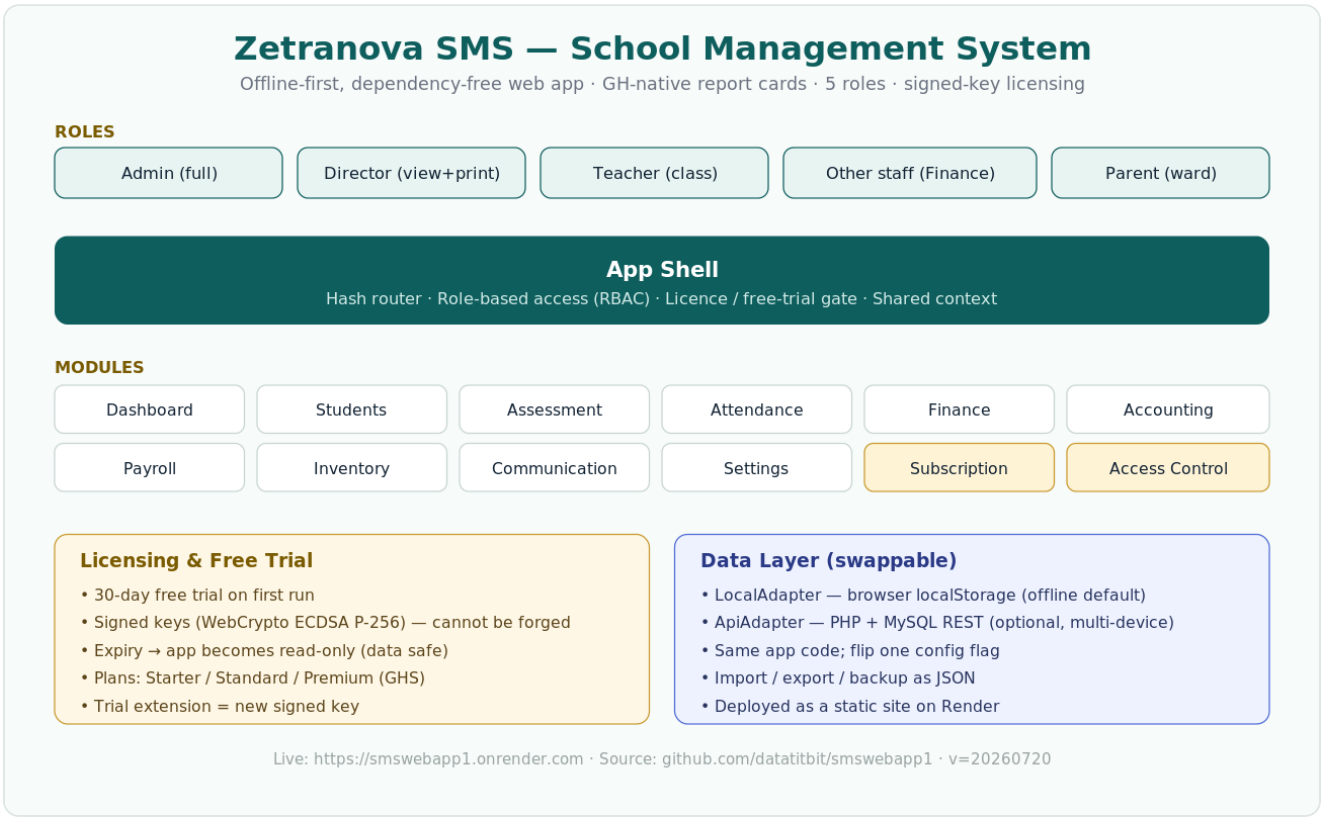
1. Overview & design philosophy

Zetranova SMS is a school-management web app for Ghanaian private schools (creche → JHS/SHS). Its design philosophy is **offline-first and dependency-free**: it runs entirely in the browser with no server required, so it works on low-bandwidth connections and keeps each school's data on their own device. It ships GH-native report cards, fees, attendance, payroll, accounting and inventory out of the box.

> **Glossary. SMS** = School Management System; **SPA** = single-page app; **offline-first** = works without internet; **SBA** = School-Based Assessment (class + exam weighting).

2. Architecture & data

The app is a vanilla-JavaScript single-page app. A thin **App Shell** handles routing, role-based access, a licence/trial gate and shared settings; each feature is an independent module. Data flows through a **swappable layer**: LocalAdapter (browser localStorage, the default) or ApiAdapter (PHP + MySQL) for multi-device use. It deploys as a static site.



> **Glossary.** **module** = a feature screen; **adapter** = interchangeable data backend; **static site** = files served as-is, no server code; **localStorage** = the browser's built-in storage.

3. Roles & access

Five fixed roles: **Admin** (full control), **Director** (view + print everywhere, no editing), **Teacher** (their class/subject — attendance + scores, no report download), **Other staff** (the finance role — Finance/Payroll/Inventory/Accounting), and **Parent** (read-only portal for their own ward). Admin decides who can access what under *Settings → Access Control*.

> **Glossary.** **RBAC** = role-based access control; **ward** = a parent's child/student; **Access Control** = admin screen for per-parent login and report-download rules.

4. Features / modules

Dashboard (KPIs + trends), Students & Admissions, Assessment / report cards, Attendance, Finance (fees, invoices, payments), Accounting (income/expense), Payroll (SSNIT/PAYE), Inventory (5 sections), Communication (SMS templates & announcements), Settings (everything configurable), and Subscription (licence/trial).

> **Glossary.** **KPI** = key performance indicator; **SSNIT/PAYE** = Ghana pension + income tax on payroll; **invoice** = a fee bill; **report card** = termly student result sheet.

5. Use cases

Daily: teachers mark attendance and enter scores; the front office admits students and records fee payments. **Termly:** generate report cards, run payroll, promote pupils. **Leadership:** the Director/CEO views dashboards and prints reports; parents check their ward's fees, attendance and results from home.

> **Glossary. promotion** = moving pupils to the next class at year-end; **terminal report** = end-of-term result sheet.

6. How to launch / deploy

The source of truth is GitHub (*datatitbit/smswebapp1*). It deploys automatically to **Render** as a free static site (publish folder *app/*). To go live: push to GitHub → Render redeploys in minutes → share **https://smswebapp1.onrender.com**. Version tags (*?v=YYYYMMDD*) force browsers to load the newest files.

> **Glossary. repo** = code repository on GitHub; **Render** = the hosting service; **deploy** = publish live; **cache-busting** = forcing browsers to fetch new files.

7. Customer / user manual (key workflows)

Sign in by choosing a role on the start screen. Admit students under *Students*; mark attendance under *Attendance*; enter class/exam scores under *Assessment*, then generate report cards. Record fees under *Finance*; run monthly payroll under *Payroll*; manage stock under *Inventory*. Configure school info, classes, grading, fees, roles and access under *Settings*. Export or back up all data any time from *Settings → Data*.

> **Glossary. seed/demo data** = sample data for trying the app; **bulk upload** = importing many records from a spreadsheet template.

8. Monetization & licensing

The app auto-starts a **30-day free trial** on first use. Plans (GHS): **Starter** ≤150 students @ 600/term, **Standard** ≤400 @ 1,200/term, **Premium** unlimited @ 2,000/term (annual prepay discounts). When a trial or term ends, the app becomes **read-only** until a licence key is entered under *Subscription*. Keys are cryptographically signed so they cannot be forged; extending a trial simply means issuing a new key.

> **Glossary. licence key** = a signed unlock code; **free trial** = full access for a limited period; **read-only** = view/print but no editing; **per-term** = billed each of the 3 school terms.

Part 2 — Updates

2.1 Settings → Access Control (per-parent login & report-download control)

2026-07-11 04:48 GMT

What it is. A new Admin-only screen that controls each parent's access. **In plain terms:** the head teacher can switch a specific parent's login on or off, and decide who may download the terminal report — for the whole school, a class, or one parent. **Example:** block report downloads school-wide during fee-arrears season, but allow one parent who has cleared. It was moved out of the teacher-visible area so only Admin can change it.

> **Glossary. report-download** = a parent printing/saving the result sheet; **per-class override** = a rule for one class; **inherit** = follow the school default.

2.2 Dashboard “Trends”: two sides + Week/Month/Term/Year filter

2026-07-11 04:48 GMT

What it is. The dashboard now shows two side-by-side panels — **Attendance & Enrolment** on the left and **Finance** (income vs expenses) on the right — sharing one time filter. **In plain terms:** leadership sees “how full is the school and how is attendance” next to “money in vs money out,” and can switch the

window from a week up to a year. Charts are hand-drawn (no external libraries) so they keep working offline.

> **Glossary.** **enrolment** = number of pupils; **trend** = change over time; **filter** = the Week/Month/Term/Year selector.

2.3 Monetization: offline licensing + 30-day free trial + Subscription screen

2026-07-11 04:48 GMT

What it is. The app can now sell itself. New schools get a 30-day free trial automatically; when it ends the app locks to read-only until they enter a licence key. Admin sees a *Subscription* screen with plans, status and an activation box. **In plain terms:** it turns the app into a product you can charge for, safely — keys are signed so nobody can fake one, and you can extend any school's trial by sending a new key. **Example:** a school likes it after 30 days, pays for Standard, and you send a paid key they paste to unlock.

> **Glossary.** **activation** = entering a key to unlock; **signed key** = tamper-proof unlock code; **read-only lock** = safe state that never deletes data.

2.4 Repository & local-folder alignment (renamed to smswebapp1, synced to v=20260720)

2026-07-11 04:48 GMT

What it is. The live GitHub repo and your desktop folder were brought in sync and both named *smswebapp1* (the desktop copy was older). **In plain terms:** there is now one clear, current version everywhere, avoiding confusion between an old local copy and the live site.

> **Glossary.** **repo** = GitHub code store; **sync** = make identical; **source of truth** = the one authoritative copy.